EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Centres of the three circles $x^2+y^2-4x-6y-14=0$, $x^2+y^2+2x+4y-5=0$ and $x^2+y^2-10x-16y+7=0$
(A) are the vertices of a right triangle
(B) the vertices of an isosceles triangle which is not regular
(C) vertices of a regular triangle
(D) are collinear

MCR001

2. $y-1=m_1(x-3)$ and $y-3=m_2(x-1)$ are two family of straight lines, right angled to each other. The locus of their point of intersection is
(A) $x^2+y^2-2x-6y+10=0$ (B) $x^2+y^2-4x-4y+6=0$
(C) $x^2+y^2-2x-6y+6=0$ (D) $x^2+y^2-4x-4y-6=0$

MCR002

3. Suppose that the equation of the circle having (–3, 5) and (5, –1) as end points of a diameter is $(x-a)^2+(y-b)^2=r^2$. Then $a+b+r, (r>0)$ is
(A) 8 (B) 9 (C) 10 (D) 11

MCR003

4. The area of an equilateral triangle inscribed in the circle $x^2+y^2-2x=0$ is
(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{8}$ (D) none

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MCR004

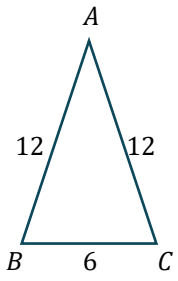
5. The smallest distance between the circle $(x-5)^2+(y+3)^2=1$ and the line $5x+12y-4=0$, is
(A) 1/13 (B) 2/13 (C) 3/15 (D) 4/15

MCR005

6. The equation of the image of the circle $x^2+y^2+16x-24y+183=0$ by the line mirror $4x+7y+13=0$ is
(A) $x^2+y^2+32x-4y+235=0$ (B) $x^2+y^2+32x+4y-235=0$
(C) $x^2+y^2+32x-4y-235=0$ (D) $x^2+y^2+32x+4y+235=0$

MCR006

7. The radius of the circle passing through the vertices of the triangle ABC, is



(A) $\frac{8\sqrt{15}}{5}$ (B) $\frac{3\sqrt{15}}{5}$ (C) $3\sqrt{5}$ (D) $3\sqrt{2}$

MCR007

For More Join: @IITJEE_Advanced

8. (6, 0), (0, 6) and (7, 7) are the vertices of a triangle. The circle inscribed in the triangle has the equation

(A) $x^2 + y^2 - 9x + 9y + 36 = 0$ (B) $x^2 + y^2 - 9x - 9y + 36 = 0$

(C) $x^2 + y^2 + 9x - 9y + 36 = 0$ (D) $x^2 + y^2 - 9x - 9y - 36 = 0$

MCR008

9. The line joining (5, 0) to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio 2 : 3 at P. If θ varies then the locus of P is :

(A) a pair of straight lines (B) a circle

(C) a straight line (D) a second degree curve which is not a circle

MCR009

10. The locus of the center of the circles such that the point (2, 3) is the mid point of the chord $5x + 2y = 16$ is

(A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$

(C) $2x + 5y + 11 = 0$ (D) none

MCR010

11. In the xy-plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is

(A) $10\sqrt{3}$ (B) $10\sqrt{5}$ (C) $10\sqrt{3} + \frac{5\pi}{3}$ (D) $10 + 5\pi$

MCR139

12. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to-

(A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 64$ (D) $x^2 + y^2 = 32$

MCR011

13. In the xy plane, the segment with end points (3, 8) and (–5, 2) is the diameter of the circle. The point $(k, 10)$ lies on the circle for

(A) no value of k (B) exactly one integral k

(C) exactly one non integral k (D) two real values of k

MCR012

14. The points (x_1, y_1), (x_2, y_2), (x_1, y_2) and (x_2, y_1) are always

(A) collinear (B) concyclic

(C) vertices of a square (D) vertices of a rhombus

MCR140

15. Locus of all point $P(x, y)$ satisfying $x^3 + y^3 + 3xy = 1$ consists of union of

(A) a line and an isolated point (B) a line pair and an isolated point

(C) a line and a circle (D) a circle and a isolated point.

MCR141

16. Consider the points $P(2, 1)$; $Q(0, 0)$; $R(4, -3)$ and the circle $S: x^2 + y^2 - 5x + 2y - 5 = 0$

(A) exactly one point lies outside S (B) exactly two points lie outside S

(C) all the three points lie outside S (D) none of the point lies outside S

MCR013

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17. The angle between the two tangents from the origin to the circle $(x-7)^2+(y+1)^2=25$ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

MCR014

18. The area of the quadrilateral formed by the tangents from the point (4, 5) to the circle $x^2+y^2-4x-2y-11=0$ with the pair of radii through the points of contact of the tangents is:

(A) 4 sq. units (B) 8 sq. units (C) 6 sq. units (D) none

MCR015

19. If a circle of constant radius $3k$ passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is -

(A) $x^2+y^2=(2k)^2$ (B) $x^2+y^2=(3k)^2$ (C) $x^2+y^2=(4k)^2$ (D) $x^2+y^2=(6k)^2$

MCR142

20. If L_1 and L_2 are the length of the tangent from (0, 5) to the circles $x^2+y^2+2x-4=0$ and $x^2+y^2-2x-y+1=0$ then

(A) $L_1=2L_2$ (B) $L_2=2L_1$ (C) $L_1=L_2$ (D) $L_1^2=L_2$

MCR016

21. From (3, 4) chords are drawn to the circle $x^2+y^2-4x=0$. The locus of the mid points of the chords is:

(A) $x^2+y^2-5x-4y+6=0$ (B) $x^2+y^2+5x-4y+6=0$

(C) $x^2+y^2-5x+4y+6=0$ (D) $x^2+y^2-5x-4y-6=0$

MCR017

22. B and C are fixed points having co-ordinates (3, 0) and (–3, 0) respectively. If the vertical angle BAC is 90°, then the locus of the centroid of the ΔABC has the equation:

(A) $x^2+y^2=1$ (B) $x^2+y^2=2$ (C) $9(x^2+y^2)=1$ (D) $9(x^2+y^2)=4$

MCR143

23. Combined equation to the pair of tangents drawn from the origin to the circle $x^2+y^2+4x+6y+9=0$ is

(A) $3(x^2+y^2)=(x+2y)^2$ (B) $2(x^2+y^2)=(3x+y)^2$

(C) $9(x^2+y^2)=(2x+3y)^2$ (D) $x^2+y^2=(2x+3y)^2$

MCR018

24. A circle of radius 5 is tangent to the line $4x-3y=18$ at $M(3,-2)$ and lies above the line. The equation of the circle, is-

(A) $x^2+y^2-6x+4y-12=0$ (B) $x^2+y^2+2x-2y-3=0$

(C) $x^2+y^2+2x-2y-23=0$ (D) $x^2+y^2+6x+4y-12=0$

MCR019

25. Tangents are drawn from (4, 4) to the circle $x^2+y^2-2x-2y-7=0$ to meet the circle at A and B. The length of the chord AB is

(A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$

MCR144

26. Tangents are drawn to a unit circle with centre at the origin from each point on the line $2x+y=4$. Then the equation to the locus of the middle point of the chord of contact is -

(A) $2(x^2+y^2)=x+y$ (B) $2(x^2+y^2)=x+2y$ (C) $4(x^2+y^2)=2x+y$ (D) none

MCR145

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For More Join: @IITJEE_Advanced

27. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point (7, 1) and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of m_1m_2, is

(A) –1 (B) 1 (C) 7/12 (D) –3

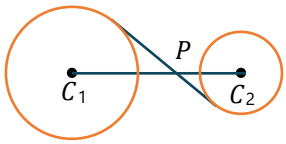
MCR146

28. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point (–1,1) and passing through the point (3,5) is-

(A) 2 (B) 3 (C) 4 (D) 5

MCR147

29. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e.$C_1C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of C_1C_2 and a common internal tangent to the circles, then $l(C_1P)$ equals-



(A) 18 (B) 20 (C) 12 (D) 15

MCR148

30. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is TANGENT to C_1 at P and to C_2 at Q is -

(A) 15 (B) 18 (C) 20 (D) 24

MCR020

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. $\frac{x-x_1}{\cos\theta}=\frac{y-y_1}{\sin\theta}=r$, represents: (Where x_1, y_1 are constant)

(A) equation of a straight line, if θ is constant and r is variable

(B) equation of a circle, if r is constant and θ is a variable

(C) a straight line passing through a fixed point and having a known slope

(D) a circle with a known centre and a given radius.

MCR049

2. Which of the following lines have the intercepts of equal lengths on the circle, $x^2+y^2-2x+4y=0$?

(A) $3x-y=0$ (B) $x+3y=0$

(C) $x+3y+10=0$ (D) $3x-y-10=0$

MCR050

3. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $x-3y+1=0$.
If two vertices of rectangle are the points (–2, 5) and (6, 5) respectively, then which of the following hold(s) good?

(A) Area of rectangle $ABCD$ is 64 square units.

(B) Centre of circle is (2, 1)

(C) The other two vertices of the rectangle are (–2, –3) and (6, –3)

(D) Equation of sides are $x=-2, y=-3, x=5$ and $y=6$.

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MCR051

4. Three concentric circles of which the biggest is $x^2+y^2=1$, have their radii in A.P. If the line $y=x+1$ cuts all the circles in real and distinct points. The permissible values of common difference of A.P. is/are

(A) 0.4 (B) 0.6 (C) 0.01 (D) 0.1

MCR052

5. The equation of circles passing through (3, –6) touching both the axes is

(A) $x^2+y^2-6x+6y+9=0$ (B) $x^2+y^2+6x-6y+9=0$

(C) $x^2+y^2+30x-30y+225=0$ (D) $x^2+y^2-30x+30y+225=0$

MCR053

6. Tangents are drawn to the circle $x^2+y^2=50$ from a point 'P' lying on the x –axis. These tangents meet the y –axis at points 'P_1' and 'P_2'. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, is/are

(A) (10, 0) (B) $(10\sqrt{2}, 0)$ (C) (–10, 0) (D) $(-10\sqrt{2}, 0)$

MCR054

7. The equations of the tangents drawn from the origin to the circle, $x^2+y^2-2rx-2hy+h^2=0$ are :

(A) $x=0$ (B) $y=0$

(C) $(h^2-r^2)x-2rhy=0$ (D) $(h^2-r^2)x+2rhy=0$

MCR055

For More Join: @IITJEE_Advanced

8. Which of the following is/are True?

The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that -

(A) they do not intersect

(B) they touch each other

(C) their exterior common tangents are parallel.

(D) their interior common tangents are perpendicular.

MCR056

9. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.

(A) Number of integral values of r are 7 for which circles are intersecting.

(B) Number of integral values of r are 9 for which circles are intersecting.

(C) For r equal to 13 number of common tangents are 3.

(D) For r equal to 21 number of common tangents are 2.

MCR057

10. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?

(A) S_1 and S_2 intersect at an angle of 90°.

(B) The point of intersection of the two circle are (2, 0) and $\left(\frac{6}{5}, \frac{8}{5}\right)$.

(C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.

(D) The point (2, 3) lies outside the circles S_1 and S_2.

MCR058

11. Two circles of radii r_1 and r_2 are both touching the coordinate axes and intersecting each other orthogonally. The value of r_1/r_2 equals -

(A) $2+\sqrt{3}$ (B) $\sqrt{3}+1$

(C) $2-\sqrt{3}$ (D) $2+\sqrt{5}$

MCR149

12. Two circles $x^2 + y^2 + px + py - 7 = 0$ and $x^2 + y^2 - 10x + 2py + 1 = 0$ intersect each other orthogonally then the value of p is -

(A) 1 (B) 2

(C) 3 (D) 5

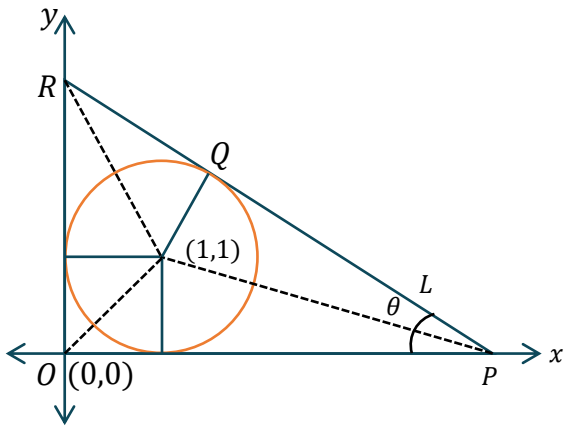
MCR150

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 13 to 15

In the diagram as shown, a circle is drawn with centre $C(1, 1)$ and radius 1 and a line L. The line L is tangential to the circle at Q. Further L meet the y –axis at R and the x –axis at P in such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.



13. The coordinates of Q are

(A) $(1 + \cos\theta, 1 + \sin\theta)$ (B) $(\sin\theta, \cos\theta)$

(C) $(1 + \sin\theta, \cos\theta)$ (D) $(1 + \sin\theta, 1 + \cos\theta)$

MCR059

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14. Equation of the line PR is

(A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$

(C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x\tan\theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$

MCR060

15. If the area bounded by the circle, the x –axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals

(A) $\sqrt{2} + 1 - \frac{3\pi}{8}$ (B) $\sqrt{2} - 1 + \frac{3\pi}{8}$ (C) $\sqrt{2} + 1 + \frac{\pi}{8}$ (D) $\sqrt{2} - 1 + \frac{\pi}{8}$

MCR061

Paragraph for Question No. 16 to 18

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, respectively then

16. Point from which length of tangents to these three circles is same is (α, β) then $(\alpha + \beta)$ is

(A) 4 (B) 5 (C) 6 (D) 7

MCR062

17. Equation of circle S_4 which cut orthogonally to all given circle is $x^2 + y^2 + ax + by + c = 0$ then $|a + b + c|$ is

(A) 16 (B) 20 (C) 24 (D) 28

MCR063

For More Join: @IITJEE_Advanced

18. Radical centre of circles $S_1, S_2, \& S_4$ is (p, q) then $|p+q|$ is

(A) 2.20 (B) 3.20 (C) 1.20 (D) 4.20

MCR064

MATRIX MATCH TYPE QUESTION

19.

	Column - I	**Column - II**
(A)	If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to	(P) 13
(B)	Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of hormonic mean between squares of their radii, then k^4 equals to	(Q) 7
(C)	The axes are translated so that the new equation of the circle $x^2+y^2-5x+2y-5=0$ has no first degree terms and the new equation $x^2+y^2=\frac{\lambda^2}{4}$, then a value of λ is	(R) 4
(D)	The number of integral points which lie on or inside the circle $x^2+y^2=4$ is	(S) 2

MCR065

20. Consider two circles C_1 of radius 'a' and C_2 of radius 'b' $(b>a)$ both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in **Column-I**, the ratio of b/a is given in **Column-II.**

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	Column-I	**Column-II**
(A)	C_1 and C_2 touch each other	(P) $2+\sqrt{2}$
(B)	C_1 and C_2 are orthogonal	(Q) 3
(C)	C_1 and C_2 intersect so that the common chord is longest	(R) $2+\sqrt{3}$
(D)	C_2 passes through the centre of C_1	(S) $3+2\sqrt{2}$
		(T) $3-2\sqrt{2}$

MCR066

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y-axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of $a + b$.

MCR151

2. If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then number of integers in the range of λ is

MCR021

3. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3. The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2. Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where m, n and p are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, find the value of $(m + n + p)$.

MCR022

4. One of the diameters of the circle circumscribing the rectangle $ABCD$ is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5,4)$ respectively, then find the area of the rectangle. (in square unit)

MCR152

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5. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0)$, $(-1, 0)$, $(1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

MCR023

6. A line with gradient 2 is passing through the point $P(1, 7)$ and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q. If (a, b) are the coordinates of the point Q, then find the value of $(7a + 7b + c)$.

MCR024

7. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2), A_3(B_3), A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5, PA_2 + PB_2 = 6, PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^{5} PA_i^2 + \sum_{i=1}^{5} PB_i^2$.

[**Note :** $A_r(B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r.]

MCR025

8. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression are $z = \frac{4 - y}{7 - x}$ M and m respectively, then find the value $(2M + 6m)$.

MCR153

For More Join: @IITJEE_Advanced

9. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at $P(6, 4)$ and both are tangent to x-axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then the value of m^2 is

MCR154

10. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab).

MCR155

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Three circles of radii $a, b, c (a < b < c)$ touch each other externally. If they have x –axis as a common tangent, then: **[JEE (Main) 2019]**

(1) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$ (2) a, b, c are in A.P.

(3) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P. (4) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$

MCR026

2. If a circle C passing through the point (4,0) touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at the point (1, –1), then the radius of C is: **[JEE (Main) 2019]**

(1) $\sqrt{57}$ (2) 4 (3) $2\sqrt{5}$ (4) 5

MCR027

3. If the area of an equilateral triangle inscribed in the circle, $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq. units then c is equal to: **[JEE (Main) 2019]**

(1) 20 (2) 25 (3) 13 (4) –25

MCR028

4. A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the coordinate axes. Then the distance of the vertex of this square which is nearest to the origin is **:-** **[JEE (Main) 2019]**

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(1) 13 (2) $\sqrt{137}$ (3) 6 (4) $\sqrt{41}$

MCR029

5. If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :- **[JEE (Main) 2019]**

(1) [12, 21] (2) (2, 17) (3) (23, 31) (4) [13, 23]

MCR031

6. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n, n \in N$, where N is the set of all natural numbers, is: **[JEE (Main) 2019]**

(1) 320 (2) 160 (3) 105 (4) 210

MCR032

7. If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q, then the locus of the mid-point of PQ is **[JEE (Main) 2019]**

(1) $x^2 + y^2 - 2xy = 0$ (2) $x^2 + y^2 - 16x^2y^2 = 0$

(3) $x^2 + y^2 - 4x^2y^2 = 0$ (4) $x^2 + y^2 - 2x^2y^2 = 0$

MCR033

8. The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point :- **[JEE (Main) 2019]**

(1) (–4, 6) (2) (6, –2) (3) (–6, 4) (4) (4, –2)

MCR034

For More Join: @IITJEE_Advanced

9. Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to: **[JEE (Main) 2020]**

(1) $\frac{52}{5}$ (2) $\frac{32}{5}$ (3) $\frac{56}{5}$ (4) $\frac{64}{5}$

MCR035

10. If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1, where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then **[JEE (Main) 2020]**

(1) $c^2 - 6c + 7 = 0$ (2) $c^2 + 6c + 7 = 0$

(3) $c^2 + 7c + 6 = 0$ (4) $c^2 - 7c + 6 = 0$

MCR036

11. If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0, (k > 0)$ touch each other at a point, then the largest value of k is _____. **[JEE (Main) 2020]**

MCR037

12. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at (2, 1), then its radius is _____ . **[JEE (Main) 2021]**

MCR038

13. If the locus of the mid-point of the line segment from the point (3, 2) to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r, then r is equal to : **[JEE (Main) 2021]**

(1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

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MCR039

14. Let the normals at all the points on a given curve pass through a fixed point (a, b). If the curve passes through (3, –3) and $\left(4, -2\sqrt{2}\right)$, and given that $a - 2\sqrt{2}\,b = 3$, then $(a^2 + b^2 + ab)$ is equal to.

[JEE (Main) 2021]

MCR040

15. Let the lengths of intercepts on x –axis and y –axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, $(a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is euqal to : **[JEE (Main) 2021]**

(1) $\sqrt{11}$ (2) $\sqrt{7}$ (3) $\sqrt{6}$ (4) $\sqrt{10}$

MCR041

16. The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$x^2 + y^2 - 10x - 10y + 41 = 0$

$x^2 + y^2 - 24x - 10y + 160 = 0$ is _____ . **[JEE (Main) 2021]**

MCR042

17. Let a circle $C: (x - h)^2 + (y - k)^2 = r^2, k > 0$, touch the x –axis at (1, 0). If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h + k + r$ is equal to. **[JEE (Main) 2022]**

MCR043

For More Join: @IITJEE_Advanced

18. The set of values of k for which the circle C: $4x^2+4y^2-12x+8y+k=0$ lies inside the fourth quadrant and the point $\left(1,-\frac{1}{3}\right)$ lies on or inside the circle C is : **[JEE (Main) 2022]**

(1) An empty set (2) $\left(6,\frac{95}{9}\right]$ (3) $\left[\frac{80}{9},10\right)$ (4) $\left(9,\frac{92}{9}\right]$

MCR044

19. A rectangle R with end points of the one of its sides as $(1, 2)$ and $(3, 6)$ is inscribed in a circle. If the equation of a diameter of the circle is $2x - y + 4 = 0$, then the area of R is _____.

[JEE(Main)-2022]

MCR045

20. Let the tangent to the circle $C_1: x^2 + y^2 = 2$ at the point $M(-1, 1)$ intersect the circle $C_2: (x - 3)^2 + (y - 2)^2 = 5$, at two distinct points A and B. If the tangents to C_2 at the points A and B intersect at N, then the area of the triangle ANB is equal to : **[JEE (Main) 2022]**

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{6}$ (4) $\frac{5}{3}$

MCR046

21. Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is **[JEE (Main) 2022]**

(1) 12 (2) 13 (3) 14 (4) 16

MCR047

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22. Points $P(-3, 2)$, $Q(9, 10)$ and $R(\alpha, 4)$ lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point S. If S lies on the line $2x - ky = 1$, then k is equal to ____.

[JEE (Main) 2023]

MCR157

23. The set of all values of a^2 for which the line $x + y = 0$ bisects two distinct chords drawn from a point $P\left(\frac{1+a}{2},\frac{1-a}{2}\right)$ on the circle $2x^2 + 2y^2 - (1 + a)x - (1 - a)y = 0$ is equal to: **[JEE (Main) 2023]**

(1) $(8,\infty)$ (2) $(4,\infty)$ (3) $(0,4]$ (4) $(2,12]$

MCR158

24. If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4},2\right)$, then the area of the triangle PQR is **[JEE (Main) 2023]**

(1) $\frac{13}{4}$ (2) $\frac{13}{8}$ (3) $\frac{5}{4}$ (4) $\frac{5}{8}$

MCR048

25. Two circles in the first quadrant of radii r_1 and r_2 touch the coordinate axes. Each of them cuts off an intercept of 2 units with the line $x + y = 2$. Then $r_2^1 + r_2^2 - r_1 r_2$ is equal to_____.

[JEE (Main) 2023]

MCR159

For More Join: @IITJEE_Advanced

26. Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x + y = 2$ intersects the circle C at the points P and Q. Let MN be a chord of C of length 2 unit and slope –1. Then, a distance (in units) between the chord PQ and the chord MN is **[JEE (Main) 2024]**

(1) $2-\sqrt{3}$ (2) $3-\sqrt{2}$ (3) $\sqrt{2}-1$ (4) $\sqrt{2}+1$

MCR160

27. A square is inscribed in the circle $x^2 + y^2 - 10x - 6y + 30 = 0$. One side of this square is parallel to $y = x + 3$. If (x_i, y_i) are the vertices of the square, then $\sum\left(x_i^2 + y_i^2\right)$ is equal to :

[JEE (Main) 2024]

(1) 148 (2) 156 (3) 160 (4) 152

MCR161

28. Let the circles $C_1 : (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2 : (x - 8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point (6, 6). If the point (6, 6) divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio 2 : 1, then $(\alpha + \beta) + 4\left(r_1^2 + r_2^2\right)$ equals **[JEE (Main) 2024]**

(1) 110 (2) 130 (3) 125 (4) 145

MCR162

29. Let the centre of a circle, passing through the point (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$, be (h, k). Then for all possible values of the coordinates of the centre (h, k), $4(h^2 + k^2)$ is equal to.

[JEE (Main) 2024]

MCR163

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30. Consider two circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - \alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii (one to each circle) drawn from one of the intersection points of C_1 and C_2 be $\sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$. If the length of common chord of C_1 and C_2 is β, then the value of $(\alpha\beta)^2$ equals.

[JEE (Main) 2024]

MCR164

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Circle(s) touching x –axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y-axis is (are) **[JEE (Advanced) 2013]**

(A) $x^2+y^2-6x+8y+9=0$ (B) $x^2+y^2-6x+7y+9=0$

(C) $x^2+y^2-6x-8y+9=0$ (D) $x^2+y^2-6x-7y+9=0$

MCR067

2. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x-1)^2+y^2=16$ and $x^2+y^2=1$. Then:- **[JEE (Advanced) 2014]**

(A) radius of S is 8 (B) radius of S is 7

(C) centre of S is (–7, 1) (D) centre is S is (–8, 1)

MCR068

3. Let RS be the diameter of the circle $x^2+y^2=1$, where S is the point (1,0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. then the locus of E passes through the point(s)- **[JEE (Advanced) 2016]**

(A) $\left(\frac{1}{3},\frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4},\frac{1}{2}\right)$

(C) $\left(\frac{1}{3},-\frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4},-\frac{1}{2}\right)$

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MCR069

4. For how many values of p, the circle $x^2+y^2+2x+4y-p=0$ and the coordinate axes have exactly three common points? **[JEE (Advanced) 2017]**

MCR071

Paragraph (Q.5 & 6)

Let S be the circle in the xy-plane defined by the equation $x^2+y^2=4$.

(There are two question based on Paragraph "X", the question given below is one of them)

5. Let E_1E_2 and F_1F_2 be the chord of S passing through the point $P_0(1,1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope –1. Let the tangents to S at E_1 and E_2 meet at E_3, the tangents of S at F_1 and F_2 meet at F_3, and the tangents to S at G_1 and G_2 meet at G_3. Then, the points E_3, F_3 and G_3 lie on the curve **[JEE (Advanced) 2018]**

(A) $x+y=4$ (B) $(x-4)^2+(y-4)^2=16$

(C) $(x-4)(y-4)=4$ (D) $xy=4$

MCR072

6. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve - **[JEE (Advanced) 2018]**

(A) $(x+y)^2=3xy$ (B) $x^{\frac{2}{3}}+y^{\frac{2}{3}}=2^{\frac{4}{3}}$

(C) $x^2+y^2=2xy$ (D) $x^2+y^2=x^2y^2$

MCR073

For More Join: @IITJEE_Advanced

7. Let T be the line passing through the points $P(-2,7)$ and $Q(2,-5)$. Let F_1 be the set of all pairs of circles (S_1,S_2) such that T is tangents to S_1 at P and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1,S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1,1)$ be F_2. Let E_2 be the set of the mid-points of the line segments in the set F_2. Then, which of the following statement(s) is (are) **TRUE**? **[JEE (Advanced) 2018]**

(A) The point $(-2,7)$ lies in E_1

(B) The point $\left(\frac{4}{5},\frac{7}{5}\right)$ does **NOT** lie in E_2

(C) The point $\left(\frac{1}{2},1\right)$ lies in E_2

(D) The point $\left(0,\frac{3}{2}\right)$ does **NOT** lie in E_1

MCR074

8. A line $y=mx+1$ intersects the circle $(x-3)^2+(y+2)^2=25$ at the points P and Q. If the midpoint of the line segment PQ has x-coordinate $-\frac{3}{5}$, then which one of the following options is correct? **[JEE (Advanced) 2019]**

(A) $6 \le m < 8$ (B) $2 \le m < 4$ (C) $4 \le m < 6$ (D) $-3 \le m < -1$

MCR075

9. Let the point B be the reflection of the point $A(2,3)$ with respect to the line $8x-6y-23=0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the line segment AC is **[JEE (Advanced) 2019]**

MCR076

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For (Q.10 to 11) Answer the following by appropriately matching the lists based on the information given in the paragraph

Let the circles $C_1: x^2+y^2=9$ and $C_2: (x-3)^2+(y-4)^2=16$, intersect at the points X and Y. Suppose that another circle $C_3: (x-h)^2+(y-k)^2=r^2$ satisfies the following conditions:

(i) centre of C_3 is collinear with the centres of C_1 and C_2

(ii) C_1 and C_2 both lie inside C_3, and

(iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2=8\alpha y$.

There are some expression given in the List-I whose values are given in List-II below :

	List-I		**List-II**
(I)	$2h+k$	(P)	6
(II)	$\frac{\text{Length of } ZW}{\text{Length of } XY}$	(Q)	$\sqrt{6}$
(III)	$\frac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$	(R)	$\frac{5}{4}$
(IV)	α	(S)	$\frac{21}{5}$
		(T)	$2\sqrt{6}$
		(U)	$\frac{10}{3}$

For More Join: @IITJEE_Advanced

10. Which of the following is the only INCORRECT combination? **[JEE (Advanced) 2019]**

(A) (IV), (S) (B) (IV), (U)

(C) (III), (R) (D) (I), (P)

MCR077

11. Which of the following is the only CORRECT combination? **[JEE (Advanced) 2019]**

(A) (II), (T) (B) (I), (S)

(C) (I), (U) (D) (II), (Q)

MCR078

12. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \frac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is **[JEE (Advanced) 2020]**

MCR079

13. Consider a triangle Δ whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is $(1, 1)$, then the equation of the circle passing through the vertices of the triangle Δ is **[JEE (Advanced) 2021]**

(A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$

(C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

MCR080

Paragraph (Q.14 & 15)

Let $M = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 \le r^2\}$, where r > 0. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}$, $n = 1, 2, 3, \ldots$. Let $S_0 = 0$ and, for $n \ge 1$, let S_n denote the sum of the first n terms of this progression. For $n \ge 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n, and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n.

14. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M. Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then **[JEE (Advanced) 2021]**

(A) $k + 2l = 22$ (B) $2k + l = 26$

(C) $2k + 3l = 34$ (D) $3k + 2l = 40$

MCR081

15. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is **[JEE (Advanced) 2021]**

(A) 198 (B) 199 (C) 200 (D) 201

MCR082

16. Let ABC be the triangle with $AB = 1, AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is _____. **[JEE (Advanced) 2022]**

MCR083

For More Join: @IITJEE_Advanced

17. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \ldots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \ldots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \ldots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE? **[JEE (Advanced) 2022]**

(A) If $n = 4$, then $(\sqrt{2}-1)r < R$
(B) If $n = 5$, then $r < R$
(C) If $n = 8$, then $(\sqrt{2}-1)r < R$
(D) If $n = 12$, then $\sqrt{2}\,(\sqrt{3}+1)r > R$

MCR084

18. Let C_1 be the circle of radius 1 with center at the origin. Let C_2 be the circle of radius r with center at the point $A = (4,1)$, where $1 < r < 3$. Two distinct common tangents PQ and ST of C_1 and C_2 are drawn. The tangent PQ touches C_1 at P and C_2 at Q. The tangent ST touches C_1 at S and C_2 at T. Mid points of the line segments PQ and ST are joined to form a line which meets the x-axis at a point B. If $AB = \sqrt{5}$, then the value of r^2 is **[JEE (Advanced) 2023]**

MCR085

19. Let the straight line $y = 2x$ touch a circle with center $(0, \alpha)$, $\alpha > 0$, and radius r at a point A_1. Let B_1 be the point on the circle such that the line segment A_1B_1 is a diameter of the circle. Let $\alpha + r = 5 + \sqrt{5}$.

Match each entry in **List-I** to the correct entry in **List-II**.

List-I		List-II	
(P)	α equals	(1)	(–2, 4)
(Q)	r equals	(2)	$\sqrt{5}$
(R)	A_1 equals	(3)	(–2, 6)
(S)	B_1 equals	(4)	5
		(5)	(2, 4)

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The correct option is **[JEE (Advanced) 2024]**

(A) (P) → (4) (Q) → (2) (R) → (1) (S) → (3)
(B) (P) → (2) (Q) → (4) (R) → (1) (S) → (3)
(C) (P) → (4) (Q) → (2) (R) → (5) (S) → (3)
(D) (P) → (2) (Q) → (4) (R) → (3) (S) → (5)

MCR165

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	A	B	D	A	B	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	B	A	D	C	B	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	C	B	C	A	C	B	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,C,D	A,B,C	C,D	A,D	A,C	A,C	A,C,D	A,C	A,C,D
Que.	11	12	13	14	15	16	17	18		
Ans.	A,C	B,C	D	C	A	B	C	A		
Que.	19			20						
Ans.	A→S;B→R;C→Q;D→P			A→S;B→R;C→Q;D→P						

EXERCISE (S)

1. 10 **2.** 4 **3.** 19 **4.** 32 **5.** 63
6. 4 **7.** 215 **8.** 4.00 **9.** 3.00 **10.** 625.00

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	2	4	1	4	3	2	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	36	3	2	9	3	1	7	4	16	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	1	4	7	2	4	2	9	1575

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C	A,C	2.00	A	D	B,D	B	10.00	A
Que.	11	12	13	14	15	16		17	18	19
Ans.	D	2.00	B	D	B	0.83 or 0.84		C,D	2.00	C

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Important Notes

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